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Document Title:		RESEARCH PROPOSAL	

(1) Research Program/Project/Study Title: Acceptability of Pandesal Enhanced with Fresh Chili Leaves (<i>Capsicum frutescens</i>) and Kulitis leaves (Amaranth)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Sigma	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☒ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Food Science	

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(9) Rationale

The popularity of baked goods is very high in the Philippines. It is one of the most well-known industries in the nation and a significant employer of labor. Bread is a widespread baked good. Pan de Sal, which translates to "salt bread" in Spanish, is one variety of famous yeast-raised bread in the Philippines. "Pan de sal" is slightly sweet rather than salty, despite its name. A bread roll produced with flour, an egg, yeast, sugar, and salt is what it is. A bread roll is a small, typically oblong or round, individual loaf of bread served as a side dish to a meal (Weller, D. 2020).

The most common bread in the Philippines is Pandesal or salt bread. It is typically served for breakfast in a plain roll or accompanied by a variety of fillings, hot cocoa, coffee, or tea. The main ingredients are flour, yeast, egg, and milk. Three hundred thirty calories, 4 grams of fat, 63 grams of total carbs, and 10 grams of protein are typically found in 100 grams of salt bread (FNRI, 1997). Most Filipinos of all ages consume salt bread every day, making it the ideal replenishment food. It has been demonstrated that adding nutrient-dense native vegetables, like squash and green moringa, to bread can significantly improve its nutritional profile (ii, iv, vi).

"Pan de sal," which was first made in the Philippines in the 16th century, is now a staple of the national breakfast. It is typically served hot and eaten by dipping it in coffee, hot cocoa, or milk. Butter, margarine, cheese, jam, peanut butter, or chocolate spread can also be added to Pandesal (<https://lilen.wikipedia.org/wiki/pandesal>). Malunggay leaves were a primary ingredient in traditional Filipino bakeries in various regions. It is referred to as "Malunggay pandesal."

Therefore, the researchers will investigate if salt bread may be improved nutritionally by using chili pepper leaves as a low-cost ingredient. The effects of crushed and powdered chili pepper leaves on the salt bread on quality, sensory attributes, and customer acceptability were evaluated.

The researchers supplemented Pandesal with fresh chili, and kulitis leaves to make it more nutrient-rich bread. In this study, the primary components of Pandesal were supplemented and enriched with fresh chili and kulitis leaves.

(9-A) Project Description/ Introduction

Filipinos have a passion for trying new foods. Filipinos think eating is the most excellent way to cope with stress and other issues. Street food is one of the most well-known cuisines among Filipinos. As a result, the researchers came up with the concept of starting a company with a significant market demand that is simple to accept. Many people in impoverished nations have inadequate diets for micronutrients like iron, vitamin A, and folate.

Iron and vitamin A deficiency continue to be widespread health issues in the Philippines among children and women (pregnant and lactating) (FNRI, 2013). Another growing issue is a folic acid deficiency, which is linked to birth abnormalities, colon cancer, and long-term physical and intellectual problems (de Vogel et al., 2008; Fujimori et al., 2011). Green leafy vegetables are extremely rich in vital micronutrients like iron, folate, and vitamin A (in the form of -carotene) (Ng et al., 2012; Gupta and Prakash, 2011). Due to their high micronutrient content, adding them as ingredients to meals or dishes is one strategy to guarantee the supply of micronutrients in the diet.

Amaranth (Kulitis) leaves have been shown to be rich in carbs, proteins, minerals, and vitamins, and regular consumption can help with weight management, improving digestion, and controlling heavy periods. It is beneficial for anemic individuals and lowers cholesterol and the risk of cardiovascular illnesses due to its high iron level and dietary fiber content (Borah, P., 2018). Abilgos-Ramos, R. et. al. (2015), claims as they examined the impact of chili pepper leaves (CPL) on salt bread's sensory properties, microbiological load, consumer acceptance, and nutritional value. According to the findings, adding CPL to salt bread can help those who live in areas with a high risk of micronutrient malnutrition get more iron, vitamin A, and folate.

The researchers became interested in conducting this investigation due to the concept mentioned earlier. The researchers wanted to create nutrient-rich Pandesal. They also want to know whether Pandesal boosted with fresh chili and kulitis leaves is palatable in terms of appearance, aroma, taste, and texture.

The utilization of chili leaves and amaranth leaves, which are readily available and highly beneficial due to the nutritional value, affordability, availability and benefit of the enhancement is abundant in the locality, this study is conducted to supplement the Filipino's access to cheaper but nutritious foods and supplement its daily nutritional value intake.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study aimed to contribute to food product development using or applying the current technology.

Specifically, it aimed to;

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1. Make Pandesal enhanced with fresh chili and kulitis leaves.
2. Determine the sensory qualities enhanced with fresh chili and kulitis leaves in terms of appearance, aroma, taste, and texture.
3. Determine the acceptability of Pandesal enhanced with fresh chili and kulitis leaves in terms of appearance, aroma, taste, and texture.

(11) Scientific Basis/Theoretical Framework

Input

Process

Output

Ingrédients, Materials
and Tools,
Equipments

Preparing
Developing
Designing

Pandesal Enhanced
with fresh Chili and
Kulitis Leaves

(12) Review of Literature/Prior Art Search

The most well-known local bread in the Philippines is Pandesal. Most bakeries across the nation, from modest backyard operations to large industrial bakeries, create and market this bread. In place of rice, this important meal is now a part of the usual Filipino breakfast. When the Spanish colonists arrived, they brought wheat flour to the islands and baked bread in wood-fired pugon ovens, giving their bread distinctive aromas and textures. In the past, pan de sal was either bought or sold by street vendors who gave out a comforting, sweet scent that could be smelled down the block.

As a result of the shortage of rice brought on by embargoes and typhoons, pan de sal gained popularity in the Manila area during World War II. According to Natalie Stein (2018), an average 45-gram serving of pan de sal has more than 100 calories. Four grams of protein, or 8% of the daily recommended intake for a healthy adult eating 2,000 calories, may be found in the bread. Pan de sal has 3 grams of total fat, or 5% of the daily recommended fat intake, and 25 grams of carbohydrates, including 1 gram of fiber. That fat contains one saturated gram. Each serving of pan de sal has 210 mg of sodium. Additionally, it includes 8% of the daily required iron intake.

A pan de sal can be taken at any time of the day; it is not simply for breakfast. Additionally, there are numerous ways to enjoy a pan de sal. Others might slice it in half and fill it with butter, sugar, or jams instead of spreading margarine on top.

As a replacement for rice, this basic dish is now a part of the typical Filipino breakfast. One cup of rice is equal to four pandesal pieces. Customers prefer their Pandesal served hot and just out of the oven. They anticipate a light brown crust with a soft crumb feel and a little crispy outside. Pandesal is typically eaten in the Philippines by dipping it into hot coffee. Additionally, several spreads or fillings, like butter, cheese, or jam, may be served with it. Today, several bakeries serve Pandesal as a snack any time of the day in addition to doing it at breakfast.

(13) Methodology

Fifteen (15) evaluators will be requested to evaluate the product. They will be composed of five (5) faculty with a recognizable background in Culinary Arts and Food Technology, five (5) members of the community picked at random, and five (5) students taking up Food Service Management as their field of specialization. They will be asked to accomplish the scorecard or evaluation sheet employing the Nine-point Hedonic Scale. After that, the data will be tabulated, analyzed, and interpreted.

Mean and One-Way Analysis of Variance (ANOVA) will be the statistical tools to analyze and interpret the data. Mean will be used to determine the sensory qualities and general acceptability of Pandesal enhanced with chili and kulitis powdered leaves in terms of appearance, aroma, taste, and texture. In contrast, Analysis of Variance (ANOVA) will be used to analyze if there were significant differences among the five treatments.

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Ingredients

The ingredients used in making Pandesal enhanced with powdered leaves of sweet potato and amaranth were: flour, brown sugar, skimmed milk, lard, rock salt, bread crumbs, unsalted butter, yeast, vegetable oil, water, fresh chili, and kulitis leaves. The experimental method will be used in the study is the Single Factor experiment with a Completely Randomized Design (CRD). It has five (5) treatments with three (3) replication each.

The treatments are as follows:

- Treatment A- (100% flour)
- Treatment B- (25% flour with 75% fresh chili and kulitis leaves)
- Treatment C- (50% flour with 50% fresh chili and kulitis leaves)
- Treatment D- (25% flour with 75% fresh chili and kulitis leaves)
- Treatment E- (100% mixture of fresh chili and kulitis leaves)

Experimental Procedure

Procedure in obtaining fresh chili and kulitis leaves

- ✓ Wash and select good quality fresh chili and kulitis leaves. Chop and then transfer it to a clear container.

Procedure in making Pandesal enhanced fresh chili and kulitis leaves

- ✓ In a spiral mixer, combine all dry ingredients (brown sugar, first-class flour, skimmed milk, yeast, and rock salt). Then, blend it into liquid ingredients (lard, butter, vegetable oil, and water). Switch on the spiral mixer at high speed to form a dough for 10 minutes until all ingredients are well mixed. Cut the dough into a quarter using a bread or dough cutter. Put the desired portion of chopped fresh chili, and kulitis leaves in a dough mixture. Knead the dough. To test whether the dough is ready, pinch a piece off and stretch it into a translucent sheet. Cut the dough lengthwise, then roll each piece into breadcrumbs. Let it rest for about 20 minutes. Cut dough into a spoonful size using a dough cutter, then arrange it individually into the aluminum baking tray. Let it rest again for 3 hours. Bake the dough for 15 minutes until golden brown.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

Activities	May 2021	June 2021	July – Aug 2021	Sep 2021	Oct 2021	Nov 2021	Dec 2021
Proposal/ Presentation							
Preparation of manuscript and Instrument							
Gathering of Data							
Data Encoding							
Interpretation of Data							
Presentation of Completed Research							

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Pandesal enhanced with Fresh Chilli (*Capsicum frutescens*) Kulitis leaves (Amaranth)

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Pandesal enhanced with Fresh Chilli (*Capsicum frutescens*) Kulitis leaves (Amaranth)

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(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

Children, teens, and adults in the community

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Prof. Jenny B. Murguia	Leader	Hotel and Restaurant Management
Dr. Leizl S. Alfonso	Member	Hospitality and Business Management
Jhonnell J. Abalajon Linelyn G. Esteban	Member Member	Hotel and Restaurant Management Hotel and Restaurant Management

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		2,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		2,000.00
- Ingredients		3,000.00
- Professional fees (laboratory analysis and other related fees)		10,000.00
- Miscellaneous (for incidental expenses)		2,000.00
- Printing		1,000.00
C. CAPITAL OUTLAY		
Total		20,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

(20) Capsule Curriculum Vitae (One-page brief C.V. only)